Assignment-3

COMPARISON OF SIMILARITY METRICS FOR TEXTURE IMAGE RETRIEVAL.

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Problem Description - The Challenge of Image Retrieval

- Rapid growth of digital image databases demands efficient retrieval methods.
- Traditional keyword-based retrieval is time-consuming and inadequate for visual content.
- Content-Based Image Retrieval (CBIR) focuses on visual features like texture.
- While texture features can be extracted effectively,
  - Similarity measurement between images is still a major challenge.
- Most systems use standard distance metrics (e.g., Euclidean) without assessing their impact on retrieval accuracy.

Paper's Proposed Solution

Feature Extraction Using Gabor Wavelets:

Applied a Gabor filter bank with:

- 4 scales × 6 orientations = 24 filters

For each filtered image:

- Calculated mean and standard deviation
- Resulting in a 48-dimensional feature vector per image.

Similarity Metrics Compared (9 Total):

- Euclidean (L2)
- Manhattan (L1)
- Mahalanobis
- Chebyshev
- Weighted-Mean-Variance (WMV)
- Canberra
- Bray-Curtis
- Squared Chord
- Squared Chi-Squared

Texture Retrieval Procedure (Original Paper)

1. Database Preparation
- Used 116 textures (Brodatz + USC + 1 artificial)
- Each 512×512 image split into sixteen 128×128 patches → 1856 images

2. Feature Extraction
- Applied Gabor filters (4 scales × 6 orientations = 24)
- Computed mean & standard deviation for each → 48D feature vector

3. Feature Database Creation

Stored 48D vectors of all 1856 images in a feature database

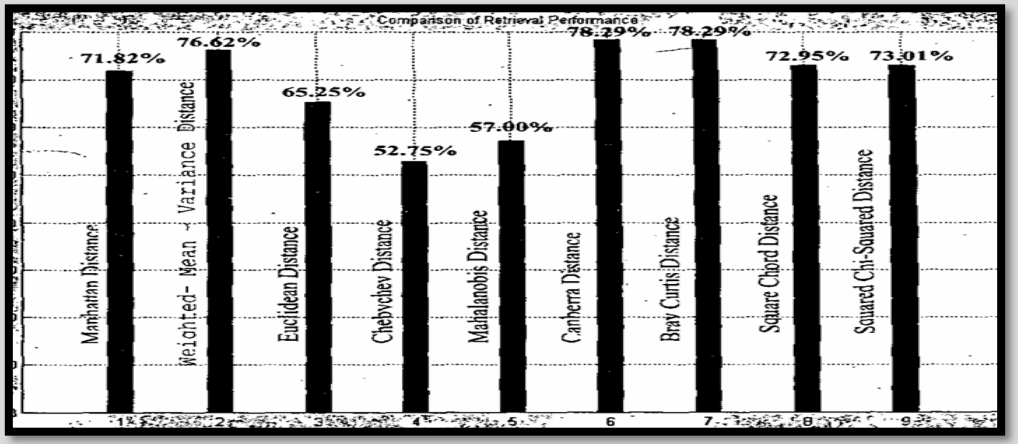
4. Query Processing

Query image also processed into a 48D vector (same pipeline)

5. Similarity Measurement
- Compared query vs. database using 9 distance metrics(e.g., Euclidean, Mahalanobis, Canberra, Bray-Curtis)
- Explored which metric yields best retrieval accuracy

6. Ranking & Retrieval
- Ranked database images by distance (smaller = more similar)
- Top matches retrieved (ideally all 16 from the same original texture)

7. Experimental Evaluation
- Implemented in MATLAB
- Found **Canberra and Bray-Curtis** gave highest retrieval accuracy



Gabor Wavelets

2D Gabor Function

- Gabor function form a complete but non-orthogonal basis set.

$$g(x,y)=\left(\frac{1}{2\pi\sigma_x\sigma_y}\right)\exp\left[-\frac{1}{2}\left[\frac{x^2}{\sigma^2{}_x}+\frac{y^2}{\sigma^2{}_y}\right]+2\pi j W_x\right]$$

where $\sigma_u = 1/2\pi\sigma_x$ and $\sigma_v = 1/2\pi\sigma_y$.

What is a Gabor Filter Bank?

- Instead of a single filter, a "Gabor filter bank" (also referred to as a "self-similar filter dictionary") is used. This bank consists of multiple Gabor filters, each tuned to different frequencies and orientations.
- This "self-similar filter dictionary" is generated by applying appropriate dilation (for scales) and rotations (for orientations) to a single "mother Gabor wavelet," g(x,y).
  - Dilation: A scale factor, a^-m, is applied to control the filter's size and frequency response, ensuring that the energy of the filter is independent of its scale m.
  - Rotation: Rotations are defined by an angle θ = nπ/K, where K is the total number of orientations

$$g_{mn}(x,y)=a^{-m}\,G(x,y), \quad a>1, \quad m,n = \text{intege}$$
$$x'=a^{-m}(x\cos\theta+y\sin\theta),$$
$$y'=a^{-m}(-x\sin\theta+y\cos\theta), \qquad (:$$

SIMILARITY MEASURES

- The distance metric can be termed as similarity measure, which is the key component in content-based image retrieval. It is important to explore different similarity measures to find the best distance metric for content based image retrieval.

1. The Euclidean or L2 metric is

$$d_E(x,y)=\sqrt{\sum_{i=1}^{d}(x_i-y_i)^2}$$

2. The Manhattan or L1 Metric is

$$d_M(x,y)=\sum_{i=1}^{d}|x_i-y_i|$$

3. Weighted- Mean-Variance

$$d_{Wmv}(x,y)=\sum_m\sum_n d_{mn}(x,y),$$

$$\text{where } d_{mn}(x,y)=\left|\frac{\mu^x{}_{mn}-\mu^y{}_{mn}}{\sigma(\mu_{mn})}\right|+\left|\frac{\sigma^x{}_{mn}-\sigma^y{}_{mn}}{\sigma(\mu_{mn})}\right|$$

4. Mahalanobis Metric

$$d_{Mah}(x,y)=\sqrt{(x-y)'\,Cov(x)^{-1}(x-y)}$$

5. Canberra Metric

$$d_c(x,y)=\sum_{i=1}^{d}\frac{|x_i-y_i|}{|x_i|+|y_i|}$$

6. Bray Curtis metric

$$d_{BC}(x,y)=\sum_{i=1}^{d}\frac{|x_i-y_i|}{x_i+y_i}$$

7 & 8. Squared Chord and Square Chi-Squared distance

$$d_{sc}(x,y)=\sum_{i=1}^{d}\left(\sqrt{x_i}-\sqrt{y_i}\right)^2$$

$$d_{chi}(x,y)=\sum_{i=1}^{d}\frac{(x_i-y_i)^2}{x_i+y_i}$$

9. Chebychev

$$D_{Cheb}(x,y)=\max_i(|x_i-y_i|)$$

Code Implementation

1. **Image Database Creation and Preprocessing**

```python
import os
import cv2
import numpy as np
import matplotlib.pyplot as plt
import math
import time
# === Configuration ===
INPUT_DIR = "C:/Users/owner/Desktop/Real_Brodatz"
OUTPUT_DIR = "C:/Users/owner/Desktop/sub img"
TARGET_IMG_SIZE = 512
SUB_IMG_SIZE = 128


# === STEP 1: Resize and Subdivide Images ===
def resize_and_split_image(img_path, save_dir, target_size=512, sub_size=128):
    img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
    if img is None:
        print(f"Skipping {img_path}: Unable to read.")
        return

    img = cv2.resize(img, (target_size, target_size), interpolation=cv2.INTER_AREA)
    base_name = os.path.splitext(os.path.basename(img_path))[0]
    idx = 0


    for i in range(0, target_size, sub_size):
        for j in range(0, target_size, sub_size):
            sub_img = img[i:i+sub_size, j:j+sub_size]
            sub_img_name = f"{base_name}_sub_{idx:02d}.png"
            sub_img_path = os.path.join(save_dir, sub_img_name)
            cv2.imwrite(sub_img_path, sub_img)
            idx += 1


def process_all_images(input_dir, output_dir):
    image_files = [f for f in os.listdir(input_dir)
                   if os.path.isfile(os.path.join(input_dir, f))]
    total = 0
    for file_name in image_files:
        img_path = os.path.join(input_dir, file_name)
        resize_and_split_image(img_path, output_dir)
        total += 16
    print(f"Finished: Generated {total} sub-images from {len(image_files)} images.")
```

2. **Texture Feature Extraction: Gabor Wavelets**

```python
# === Parameters as per paper ===
sigma_x = 2.5
sigma_y = 2.5
W_x_list = [0.05, 0.1, 0.2, 0.4]  # Frequency band values
orientations_deg = [0, 30, 60, 90, 120, 150]
kernel_size = 31
FEATURE_SAVE_PATH = "C:/Users/owner/Desktop/gabor_features.npy"

# === STEP 2: Gabor Filter Definition with Scaling ===
def gabor_function_from_paper(x, y, sigma_x, sigma_y, W_x, theta, a_power_m_factor):
    # Use fixed sigma_x and sigma_y (mother wavelet spread) passed in; do NOT recalc here

    # Rotate coordinates
    x_rot = x * np.cos(theta) + y * np.sin(theta)
    y_rot = -x * np.sin(theta) + y * np.cos(theta)

    # Gaussian envelope (no scaling of coordinates)
    exponent = -0.5 * ((x_rot**2) / (sigma_x**2) + (y_rot**2) / (sigma_y**2))

    # Gabor function with cosine modulation
    gabor = (1 / (2 * np.pi * sigma_x * sigma_y)) * np.exp(exponent) * np.cos(2 * np.pi * W_x * x_rot)

    # Apply energy normalization factor as an outer multiplier
    gabor *= a_power_m_factor

    return gabor
```

```python
def generate_gabor_kernel(sigma_x, sigma_y, W_x, theta_deg, size, a_power_m_factor):
    theta = np.deg2rad(theta_deg)
    kernel = np.zeros((size, size), dtype=np.float32)
    half = size // 2
    for i in range(-half, half + 1):
        for j in range(-half, half + 1):
            kernel[i + half, j + half] = gabor_function_from_paper(i, j, sigma_x, sigma_y, W_x, theta, a_power_m_factor)
    return kernel
```

```python
# === STEP 3: Convolution and Feature Extraction ===
def convolve_image(img, kernel):
    h, w = img.shape
    kh, kw = kernel.shape
    pad_h, pad_w = kh // 2, kw // 2
    padded = np.pad(img, ((pad_h, pad_h), (pad_w, pad_w)), mode='reflect')
    out = np.zeros_like(img, dtype=np.float32)

    for i in range(h):
        for j in range(w):
            region = padded[i:i+kh, j:j+kw]
            out[i, j] = np.sum(region * kernel)
    return out


def extract_features(img, kernel_bank):
    features = []
    for kernel in kernel_bank:
        filtered = convolve_image(img, kernel)
        features.append(np.mean(filtered))
        features.append(np.std(filtered))
    return np.array(features, dtype=np.float32)


# === STEP 4: Main Pipeline ===
def main():
    # Step 1: Preprocess images
    process_all_images(INPUT_DIR, OUTPUT_DIR)

    # Step 2: Generate Gabor kernel bank (24 kernels)
    kernel_bank = []
    for m, W_x in enumerate(W_x_list):  # m = 0 to 3
        a_power_m_factor = (2) ** (-m)  # a = 2, so a^-m
        for theta in orientations_deg:
            kernel = generate_gabor_kernel(sigma_x, sigma_y, W_x, theta, kernel_size, a_power_m_factor)
            kernel_bank.append(kernel)

    # Step 3: Extract features from each sub-image
    features = []
    file_list = sorted([f for f in os.listdir(OUTPUT_DIR) if f.endswith(('.png', '.jpg'))])
    for fname in file_list:
        path = os.path.join(OUTPUT_DIR, fname)
        img = cv2.imread(path, cv2.IMREAD_GRAYSCALE)
        if img is None:
            print(f"Skipping {fname}")
            continue
        vec = extract_features(img, kernel_bank)
        features.append(vec)
```

3. Feature Database Creation

```python
# Step 4: Save features
feature_matrix = np.array(features)
np.save(FEATURE_SAVE_PATH, feature_matrix)
print(f"Done! Extracted features for {len(features)} images. Saved at: {FEATURE_SAVE_PATH}")
```

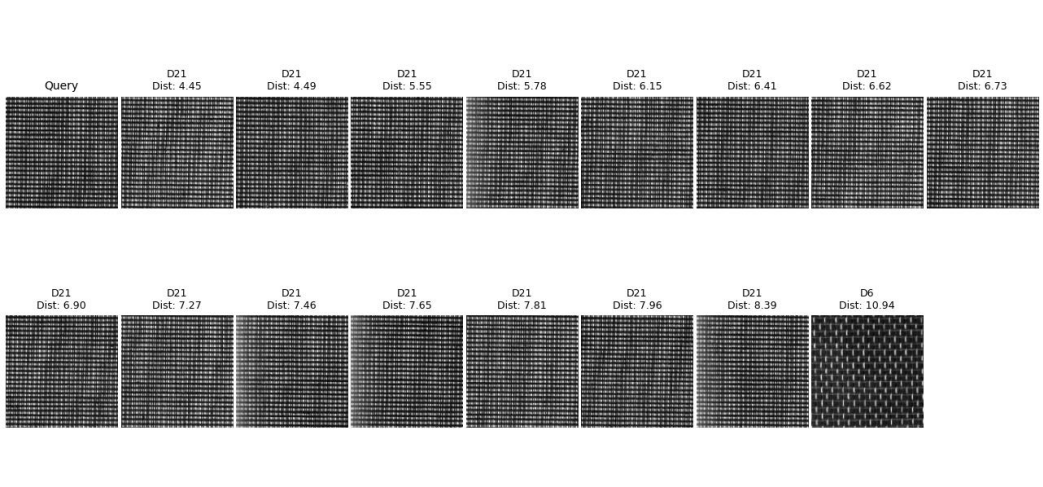
4. Similarity Measures/ Distance Metrics

```python
# === Distance Metrics ===
def manhattan_distance(x, y):
    return np.sum(np.abs(x - y))

def weighted_mean_variance_distance(x, y, std):
    return np.sum(np.abs(x - y) / std)

def euclidean_distance(x, y):
    return np.sqrt(np.sum((x - y) ** 2))

def chebyshev_distance(x, y):
    return np.max(np.abs(x - y))

def mahalanobis_distance(x, y, inv_cov):
    diff = x - y
    val = np.dot(diff, np.dot(inv_cov, diff))
    val = max(val, 0)
    return np.sqrt(val)


def canberra_distance(x, y):
    x = np.asarray(x)
    y = np.asarray(y)
    return np.sum(np.abs(x - y) / (np.abs(x) + np.abs(y)))

def bray_curtis_distance(x, y):
    x = np.asarray(x)
    y = np.asarray(y)
    return np.sum(np.abs(x - y)) / np.sum(x + y)

def squared_chord_distance(x, y):
    x = np.asarray(x)
    y = np.asarray(y)
    return np.sum((np.sqrt(x) - np.sqrt(y)) ** 2)


def squared_chi_squared_distance(x, y):
    num = (x - y) ** 2
    denom = x + y
    return np.sum(num / denom)
```

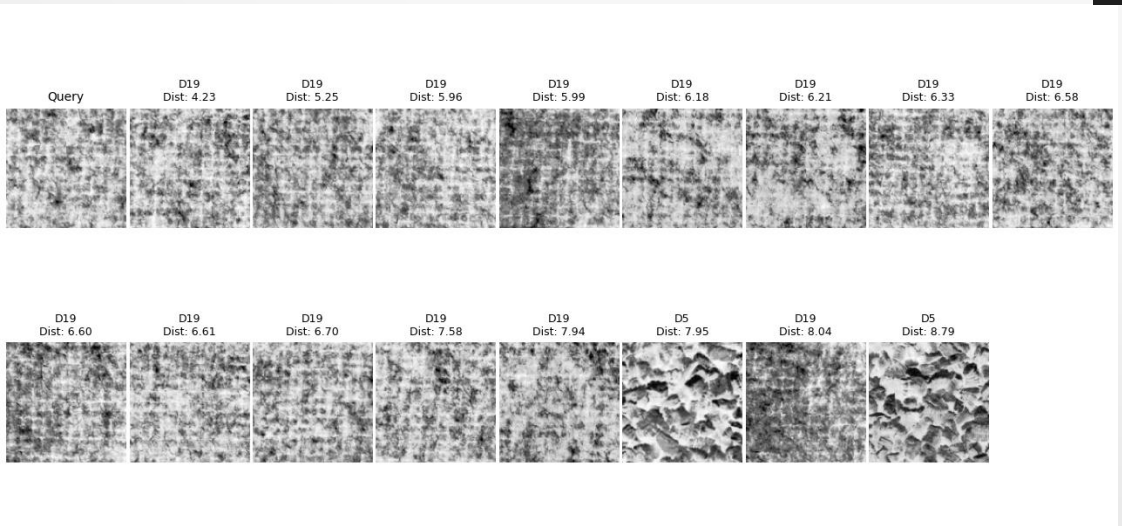
```python
def evaluate_all_queries(top_N=16):

    start_time = time.time()

    database_features = np.load(FEATURE_SAVE_PATH)
    file_list = sorted([f for f in os.listdir(OUTPUT_DIR) if f.endswith(('.png', '.jpg'))])
    num_images = len(file_list)

    std = np.std(database_features, axis=0)
    std[std == 0] = 1e-5

    cov = np.cov(database_features.T)
    inv_cov = np.linalg.pinv(cov)

    total_relevant_retrieved = {metric: 0 for metric in [
        "Manhattan", "Weighted-Mean-Variance", "Euclidean", "Chebyshev", "Mahalanobis",
        "Canberra", "Bray-Curtis", "Squared-Chord", "Squared-Chi-Squared"
    ]}
```

```python
def get_parent_class(fname):
    return fname.split('_sub_')[0]

for query_index in range(num_images):
    query_vec = database_features[query_index]
    query_name = file_list[query_index]
    query_class = get_parent_class(query_name)

    distances = {
        "Manhattan": [],
        "Weighted-Mean-Variance": [],
        "Euclidean": [],
        "Chebyshev": [],
        "Mahalanobis": [],
        "Canberra": [],
        "Bray-Curtis": [],
        "Squared-Chord": [],
        "Squared-Chi-Squared": []
    }

    for i, db_vec in enumerate(database_features):
        if i == query_index:
            continue
        db_name = file_list[i]
        distances["Manhattan"].append((db_name, manhattan_distance(query_vec, db_vec)))
        distances["Weighted-Mean-Variance"].append((db_name, weighted_mean_variance_distance(query_vec, db_vec, std)))
        distances["Euclidean"].append((db_name, euclidean_distance(query_vec, db_vec)))
        distances["Chebyshev"].append((db_name, chebyshev_distance(query_vec, db_vec)))
        distances["Mahalanobis"].append((db_name, mahalanobis_distance(query_vec, db_vec, inv_cov)))
        distances["Canberra"].append((db_name, canberra_distance(query_vec, db_vec)))
        distances["Bray-Curtis"].append((db_name, bray_curtis_distance(query_vec, db_vec)))
        distances["Squared-Chord"].append((db_name, squared_chord_distance(query_vec, db_vec)))
        distances["Squared-Chi-Squared"].append((db_name, squared_chi_squared_distance(query_vec, db_vec)))
```

Results

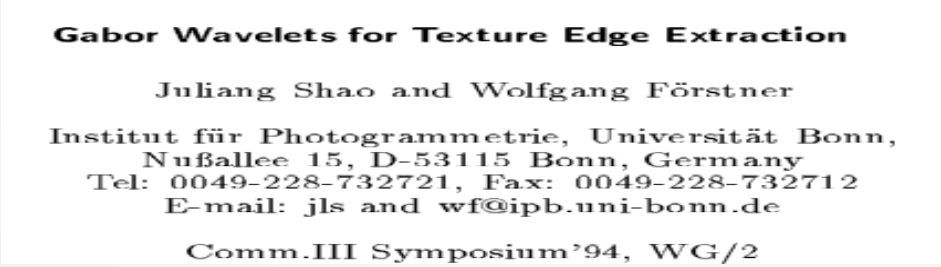



```
Average Retrieval Accuracy (Precision@Top-16):
  Manhattan: 45.06% (Time: 1.9786s)
  Weighted-Mean-Variance: 70.85% (Time: 1.9007s)
  Euclidean: 42.10% (Time: 2.0009s)
  Chebyshev: 41.54% (Time: 1.7402s)
  Mahalanobis: 34.18% (Time: 2.5438s)
  Canberra: 69.35% (Time: 2.9351s)
  Bray-Curtis: 45.17% (Time: 3.3038s)
  Squared-Chord: 3.45% (Time: 3.9733s)
  Squared-Chi-Squared: 55.83% (Time: 2.0629s)

Total Retrieval Time for All Queries: 24.0203 seconds
```



References

1. Aswini. K. Samantaray, Amol D. Rahulkar, “Comparison of Similarity Measurement Metrics on Medical Image Data”, IEEE – 45670, Available: https://ieeexplore.ieee.org/document/8944781
2. Juliang Shao and Wolfgang Förstner, “Gabor Wavelets for Texture Edge Detection“.



Gabor Wavelets for Texture Edge Extraction

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Comm.III Symposium’94, WG/2

3. Manesh Kokare, B.N. Chatterji and P.K. Biswas, “A survey on current content based image retrieval methods,” IETE Journal of Research, Vol. 48, No.3&4, 261-271, May-Aug 2002.
4. B.S. Manjunath and W.Y. Ma, “Tcxture features for browsing and retrieval of image data”, IEEE Trans. on Pattern Analysis and Machine Intelligence, 8(8), 837- 842, Aug 1996.

Thank You